

Relational Database

Mini-project

2025

I. Objective

Students will design and implement a relational database using MySQL, applying concepts of database design, normalization, and SQL queries. By the end of the project, students will be able to:

- Design a normalized relational schema.
- Implement the database using SQL in MySQL.
- Perform CRUD operations and advanced queries using SQL in MySQL.

II. Tools

- MySQL version 8.0 or later.
- MySQL Workbench (or MySQL Shell, HiedisQL, DBeaver, phpMyAdmin).
- No additional tools or programming languages are required.

Note: Application development, user interface (UI) design, or the use of programming languages (e.g., Python, Java, JavaScript) is not required. Students should focus solely on database design and SQL queries in **relational** model.

III. Topics

For choosing a topic:

- One topic can be chosen by more than one group but each team need to design a different database schema.
- Each team can expand or modify the description for their chosen topic during the requirement analysis phase of the project
- Choose one of the following topics or propose your own:

1. Hospitality and Tourism

1. **Hotel Room Booking Management:** Manage room reservations, check-ins, check-outs, and room availability.
2. **Airline Reservation System:** Manage flights, passengers, tickets, and seat availability.
3. **Car Rental System:** Manage cars, customers, rentals, and payments.

4. **Tourism and Travel Management:** Manage tour packages, customers, bookings, and payments.
5. **Event Management System:** Manage events, attendees, venues, and schedules.

2. Education and Learning

6. **Student Attendance Management:** Track student attendance, courses, and schedules.
7. **Online Course Registration System:** Manage students, courses, instructors, and enrollments.
8. **School Management System:** Manage students, teachers, classes, subjects, and grades.

3. Healthcare and Wellness

11. **Patient Information Management:** Manage patient records, appointments, and medical history.
12. **Hospital Management System:** Manage doctors, patients, appointments, and medical records.
13. **Pharmacy Management System:** Manage medicines, suppliers, customers, and prescriptions.
14. **Fitness Center Management:** Manage members, trainers, workout schedules, and attendance.
15. **Gym & Fitness Membership System:** Track memberships, attendance, and payments.

4. Retail and E-commerce

16. **Point of Sale (POS) System:** Manage sales transactions, inventory, and customer details.
17. **E-commerce Platform:** Manage products, categories, customers, orders, and payments.
18. **Clothing Store Inventory Management:** Track clothing items, sizes, stock levels, and sales.
19. **Online Bookstore:** Manage books, authors, publishers, and orders.
20. **Video Game Store:** Manage games, platforms, customers, and purchases.

5. Logistics and Inventory

21. **Freight & Logistics Management:** Manage shipments, customers, and delivery tracking.
22. **Inventory & Stock Management:** Track products, suppliers, orders, and stock levels.

23. **Farm Management System:** Manage crops, livestock, equipment, and workers.

24. **Waste Management System:** Manage waste collection, bins, vehicles, and workers.

6. Entertainment and Media

25. **Movie Ticket Booking System:** Manage movies, theaters, showtimes, and bookings.

26. **Music Streaming Platform:** Manage songs, artists, albums, and playlists.

27. **Social Media Platform:** Manage users, posts, comments, likes, and followers.

7. Finance and Banking

28. **Bank Transactions & Customer Management:** Manage accounts, transactions, and customer details.

29. **Insurance Policy & Claims Management:** Manage policies, customers, and claims.

30. **Charity Donation System:** Manage donors, donations, campaigns, and beneficiaries.

8. Automotive and Services

31. **Vehicle Service Center Management:** Manage vehicles, customers, services, and appointments.

32. **Car Information Management:** Track car details, owners, and maintenance history.

9. Food and Beverage

33. **Restaurant Order & Billing System:** Manage orders, menus, customers, and payments.

34. **Cafeteria Management System:** Manage food items, orders, customers, and payments.

35. **Online Food Delivery System:** Manage restaurants, menus, orders, and delivery personnel.

10. Others

36. **Online Auction System:** Manage auctions, bidders, items, and bids.

37. **Freelance Platform:** Manage freelancers, clients, projects, and payments.

38. **Real Estate Management System:** Manage properties, agents, buyers, and sales.

39. **Online Job Portal:** Manage job postings, applicants, employers, and resumes.

40. **Pet Adoption System:** Manage pets, shelters, adopters, and adoption requests.

41. **Laboratory Equipment Tracking:** Manage lab equipment, maintenance, and usage.

42. **Online Voting System:** Manage candidates, voters, elections, and votes.

43. **Campus Event Management:** Manage events, participants, organizers, and schedules.

IV. Tasks and Deliverables

- **Step-by-step guides:**
 - **Phase 1: Requirements Analysis**
 - Select a topic
 - Define the scope and requirements of the chosen topic.
 - Identify entities, attributes, and relationships.
 - **Phase 2: Database Design**
 - Design and Normalize the database schema (up to 3NF or BCNF).
 - Create an ER or database diagram.
 - **Phase 3: Implementation**
 - Write SQL scripts to create the database and tables.
 - Apply integrity constraints (e.g., primary keys, foreign keys, unique, check, null, default).
 - **Phase 4: CRUD Operations**
 - Write SQL to Insert sample data, retrieve, update, and delete data
 - Write SQL queries to perform operations following the requirements
 - **Deliverable 5: Advanced Queries**
 - Write complex queries using joins, subqueries, and aggregate functions.
 - Generate reports (e.g., monthly sales, top-selling products)
 - **Deliverable 6: Presentation slide and exported database**
 - Exported Database: A .sql file of the entire database schema and data.
 - SQL Script: A single .sql file containing all CRUD and advanced queries
 - A full report
 - Presentation slide

V. Evaluation Criteria

N0	Evaluation Criteria		Weight
I	Team Score		50%
1.		Team Performance and Progress	10%
2.		Database design – schema and diagram	10%
3.		Database implementation – DDL, DML and constraints	10%
4.		Advanced queries – joins, subqueries, aggregates	10%
5.		Report	10%
II	Individual Score		50%
6.		Technical knowledge	20%
7.		Contribution	20%
8.		Presentation	10%

VI. Project Deliverables

- a. **Database Diagram:** Export as PDF or PNG or link.
- b. **SQL Script:** A single .sql file containing all commands (e.g., create_tables.sql, insert_data.sql, queries.sql).
- c. **Exported Database:** A .sql dump file of the entire database.
- d. **Report:** A PDF document with the specified structure.

VII. Report Structure

- **Title Page:** Project title, course name, team number, team members, date
- **Table of Contents:** List all sections and page numbers.
- **Introduction:**
 1. Project description and objectives.
 2. Scope and requirements.
 3. Team members and assigned tasks.
- **Database Design:**
 1. Database diagram with explanation.

2. List or table of schema (tables, attributes, relationships, constraints).
- **Implementation:**
 - o SQL scripts for implementing the database, such as defining the structure and constraints, manipulating and retrieving the data.
 - o SQL scripts to show the structure of each table, including its columns, data types, and constraints
 - o SQL scripts to show the tables of constraints used in the designed database
 - o SQL scripts to CRUD data
 - o Screenshots of query results.
 - **Advanced Queries:**
 - o List of SQL queries with explanations.
 - o Screenshots of query results.
 - **Conclusion:**
 - o Limitation of the proposed database design.
 - o Challenges faced and lessons learned.
 - **References:** Any resources or tools used.

VIII. Project Timeline

- Project starts: 04/20/2026
- Project progress demonstration in TP classes 3mn per team and Submit SQL scripts for advanced queries by the **end of TP class**
 - Topic, project requirement: TP1
 - Database design and diagram: TP2
 - Database implementation and constraint in MySQL: TP3
 - Database CRUD operation: TP4
 - Advanced Queries: TP5
- Submit all deliverables: by 5/31, 11:59 PM
- Presentation + Q&A: 6/03 - 04 (TP6)
- Final Exam: 6/10-11